

Task 1: Use the systemctl command to check the status of the sshd service on your Linux system and write down whether it is active, inactive, or failed.

Ans :-

Command

```
sudo systemctl status sshd
```

Or check only the current state:

```
systemctl is-active sshd
```

Possible Output

active

or

inactive

or

failed

Explanation

- **active** → SSH service is running and accepting connections.
- **inactive** → SSH service is installed but not running.
- **failed** → SSH service attempted to start but encountered an error.

**Task 2: Start the nginx service using systemctl, then immediately stop it. After each action, verify the status and note the changes.

Hint: Use 'sudo systemctl start nginx' and 'sudo systemctl stop nginx'.**

Ans :-

Step 1: Install Nginx (if not installed)

```
sudo dnf install nginx -y
```

Step 2: Start Nginx

```
sudo systemctl start nginx
```

Verify:

```
sudo systemctl status nginx
```

Expected:

Active: active (running)

or

```
systemctl is-active nginx
```

Output:

active

Step 3: Stop Nginx

```
sudo systemctl stop nginx
```

Verify:

```
sudo systemctl status nginx
```

Expected:

Active: inactive (dead)

or

```
systemctl is-active nginx
```

Output:

inactive

Observation

Action	Status
--------	--------

After Start	active (running)
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After Stop	inactive (dead)
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Task 3: Enable the cron service to start automatically on boot, then reboot your system and confirm that the service is running after reboot.

Ans :-

On CentOS Stream 9, the cron service is **crond**.

Enable service

```
sudo systemctl enable crond
```

Expected:

Created symlink...

Start service

```
sudo systemctl start crond
```

Verify

```
sudo systemctl status crond
```

Expected:

Active: active (running)

Reboot

```
sudo reboot
```

After reboot

Check again:

```
sudo systemctl status crond
```

or

```
systemctl is-enabled crond
```

Expected:

enabled

and

active (running)

Task 4: List all currently running services using systemctl and identify any service related to networking (like NetworkManager or networking). Write the current status of that service.

Ans :-

Show running services

```
systemctl list-units --type=service --state=running
```

Example Output

NetworkManager.service

crond.service

sshd.service

systemd-journald.service

firewalld.service

Check networking service

Most CentOS Stream 9 systems use **NetworkManager**.

```
sudo systemctl status NetworkManager
```

or

```
systemctl is-active NetworkManager
```

Expected

active (running)

Observation

Example:

Service	Status
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NetworkManager	active (running)
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Task 5: Simulate a real-world scenario: Imagine you are troubleshooting why Spotify Web cannot stream music on your Linux machine. Use systemctl to check if the firewall (ufw or firewalld) service is running, and if it is, disable it temporarily. Document each command used and the result.
Constraint: Only disable the firewall if you are on a safe, non-production environment.

Ans :-

Scenario: Spotify Web is not streaming music.

Step 1: Check Firewalld

CentOS Stream 9 uses **firewalld** by default.

```
sudo systemctl status firewalld
```

or

```
systemctl is-active firewalld
```

Possible output

active

or

inactive

Step 2: Stop Firewalld (Temporary)

Only perform this on a safe lab or non-production system.

```
sudo systemctl stop firewalld
```

Step 3: Verify

```
sudo systemctl status firewalld
```

Expected

Active: inactive (dead)

or

```
systemctl is-active firewalld
```

Output

inactive

Step 4: Re-enable Firewall

After testing, restart it:

```
sudo systemctl start firewalld
```

Verify:

```
sudo systemctl status firewalld
```

Expected

Active: active (running)